DATA ITEM DESCRIPTION

Title: MACHINERY AND EQUIPMENT AIRBORNE SOUND MEASUREMENTS TEST PLAN

Number: DI-HFAC-80270A
AMSC Number: N9536
DTIC Applicable: No
Preparing Activity: SH
Applicable Forms:

Approval Date: 20150413
Limitation: N/A
GIDEP Applicable: No
Project Number: HFAC-2015-004

Use/relationship: The Machinery and Equipment Airborne Sound Measurements Test Plan describes the airborne sound measurements test procedure, facility, and measurement system in advance of the test to demonstrate planned compliance with the applicable requirements of MIL-STD-1474 and ANSI/ASA S12.67, *Pre-Installation Airborne Sound Measurements and Acceptance Criteria of Shipboard Equipment*.

a. This DID shall be applied for each noise grade, class, rating, and type of equipment to be tested for airborne noise performance when invoked on each contract or order, or when MIL-STD-1474 or ANSI/ASA S12.67 is invoked in the contract specification or statement of work (SOW).

b. This Data Item Description (DID) contains the format, content, and preparation instructions for the Machinery and Equipment Airborne Sound Measurements Test Plan in response to the applicable tasks in the contract SOW.

c. This DID supersedes DI-HFAC-80270.

Requirements:

1. Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions shall be as specified in the contract.

2. Format. The Machinery and Equipment Airborne Sound Measurements Test Plan shall be prepared in Contractor's format.

3. Content. The Machinery and Equipment Airborne Sound Measurements Test Plan shall contain the following information:

3.1. Test and equipment information. The report shall contain the following:

a. Equipment to be evaluated, including contract number.

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b. Ship name(s) and hull number(s), if known, for which the unit is being acquired.
c. Evaluation measurements to be performed.
d. Evaluating activity and test site.
e. Test date, if established.
f. Equipment specification acceptance criteria.
g. Driven and driver unit manufacturers.
h. Driver unit horsepower.
i. Rated electrical characteristics, including volts, amps, and phase.
j. Rated fluid media parameters (e.g., flow rates, temperatures, pressures).
k. Equipment rated operational speeds (e.g., driver and driven unit revolutions per minute).
l. Expected equipment noise and vibration sources and their frequencies of excitation (e.g., rotationals, gear tooth contact, blade rate).
m. Resilient mount manufacturer and model number.
n. Dates of manufacture of resilient mounts to be used.
o. Number of resilient mounts to be used.
p. If the use of non-Navy standard resilient mounts has been approved, include the characteristics including load rating in pounds and resonant frequency data of the mounts to be used.
q. Planned equipment load during test (e.g., full, half).
r. Planned operating speeds during test.
s. Planned throttling conditions, if any.
t. Planned operating suction and discharge fluid media parameters during test (e.g., flow rates, pressures, temperatures).
u. Planned electrical inputs, outputs, or both during test.
v. Block diagrams and identification of make and model of measurement and analysis instrumentation to be used.
w. Block diagrams and identification of make and model of system calibration instrumentation to be used.
x. Place and date of latest laboratory calibration of measurement instruments, system calibration instruments (including field calibrators), and analysis instruments which will be used.

3.2. Drawings, sketches, and photographs. The report shall contain the following:

a. Overall unit dimensions and orientation (e.g., vertical, horizontal) to be used during test.
b. Identification of major components.
c. Interface information.
d. Mounting locations and dimensions.
e. Weights and center of gravity for major components.
f. Resilient mount orientation (e.g., vertical, horizontal, angle), approximate load on each mount, attachment locations, overall unit weight, and center of gravity.

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g. Pedestal or subbase dimensions, materials, and weight, when applicable.
h. Dimensions, materials, and weight of the mounting fixture if a mounting fixture other than the standard fixture is to be used.
i. Planned arrangement and description of flexible pipes and ducts, if to be used.
j. Planned arrangement of test piping and fittings.
k. Planned arrangement of electrical power leads.
l. Planned location and description of test instrumentation.
m. Planned locations of resilient mounts and pedestals or fixtures that support the equipment.
n. Approximate locations and orientation of each microphone relative to equipment features.
o. Locations of equipment air inlet and discharge openings.
p. Description of acoustical center of sound source, assumed or measured.
q. Acoustical environment (indoor). When testing indoors, the following shall be included:

    (1) Description of planned location and orientation of test unit relative to floor, walls, ceiling, and contents of test room.
    (2) Description of planned test location (e.g., anechoic, reverberant); dimensions of the test room or chamber; and description of the physical treatment on walls, ceiling, and floor.

r. Acoustical environment (outdoor). When testing outdoors, the following shall be included:

    (1) Description of planned location and orientation of test unit relative to buildings, structures, or other reflecting surfaces or objects.
    (2) Description of the planned ground surface, and wind direction and speed.

3.3. Tabular and graphic formats. Description of tabular and graphic formats to be used in tabulating and recording airborne sound measurement data.

4. End of DI-HFAC-80270A.

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